PRN – 260245920034

1. *Display Name of the salesperson, customer name and amount of sales.*

```
select
  case s.repid
    when 7844 then 'john smith'
    when 7521 then 'mary johnson'
    when 7654 then 'david lee'
    when 7499 then 'susan brown'
    else 'unknown'
  end as salesperson,
  s.custname as customer,
  s.amount   as saleamount
from sales s;
+--------------+----------------+------------+
| salesperson  | customer       | saleamount |
+--------------+----------------+------------+
| john smith   | JOCKSPORTS     |    1703.40 |
| john smith   | JOCKSPORTS     |     485.00 |
| john smith   | JOCKSPORTS     |     292.50 |
| john smith   | JOCKSPORTS     |      50.00 |
| john smith   | JOCKSPORTS     |     350.00 |
| john smith   | JOCKSPORTS     |    2400.00 |
| john smith   | NORTH WOODS    |     440.00 |
+--------------+----------------+------------+
47 rows in set (0.00 sec)
```

2. *List employee name and name of managers.*

```
mysql>  select e.ename as employee,  m.ename as manager
        from emp e left join emp m
        on e.mgr = m.empno
        order by e.empno;;
+----------+---------+
| Employee | Manager |
+----------+---------+
| GITA     | GRASS   |
| VRUSHALI | GRASS   |
| SHARMIN  | GRASS   |
| BANDISH  | GRASS   |
| KING     | NULL    |
| SUPRIYA  | GRASS   |
+----------+---------+
30 rows in set (0.00 sec)
```

3. *Display details of all departments and the corresponding employees*

```
mysql>  select d.deptno, d.dname, d.loc, e.empno, e.ename, e.job, e.sal, e.hiredate
```

```
from dept d left join emp e on d.deptno = e.deptno
order by d.deptno, e.empno;

+--------+------------+----------+-------+----------+---------------------+------+------------+------+-------+
| deptno | dname      | loc      | empno | ename    | job                 | sal  | hiredate   |
+--------+------------+----------+-------+----------+---------------------+------+------------+------+-------+
|     10 | ACCOUNTING | NEW YORK |  7415 | AARAV    | CLERK               | 3350 | 1981-12-31 |
|     10 | ACCOUNTING | NEW YORK |  7421 | THOMAS   | CLERK               | 1750 | 1981-07-19 |
|     10 | ACCOUNTING | NEW YORK |  7782 | CLARK    | MANAGER             | 2450 | 1981-06-09 |
|     10 | ACCOUNTING | NEW YORK |  7839 | KING     | PRESIDENT           | 5000 | 1981-11-17 |
|     10 | ACCOUNTING | NEW YORK |  7934 | MILLER   | CLERK               | 1300 | 1982-01-23 |
|     20 | RESEARCH   | DALLAS   |  6473 | SHARMIN  | Public Relation     | 1750 | 1981-07-19 |
|     20 | RESEARCH   | DALLAS   |  7788 | SCOTT    | ANALYST             | 3000 | 1982-12-09 |
+--------+------------+----------+-------+----------+---------------------+------+------------+
31 rows in set (0.00 sec)
```

4. *Display all customer names with their order details (order number and ship date) for the orders shipped in 1987.*

```
mysql> select c.name as customer_name,
    o.onum as order_number,
    o.odate as ship_date
from customer c
join orders o
on c.custid = o.cnum
where year(o.odate) = 1987
order by c.name, o.onum;

Empty set (0.00 sec)
```

5. *Display salesman names and names of customers they are dealing with.*

```
mysql> select s.ename as salesman_name,
    c.name as customer_name
from emp s
join customer c
on s.empno = c.repid
order by s.ename, c.name;
+---------------+----------------+
| salesman_name | customer_name  |
+---------------+----------------+
| ALLEN         | EVERY MOUNTAIN |
| ALLEN         | WOMENS SPORTS  |
| MARTIN        | VOLLYRITE      |
| TURNER        | JOCKSPORTS     |
| TURNER        | K + T SPORTS   |
| TURNER        | NORTH WOODS    |
| WARD          | JUST TENNIS    |
| WARD          | SHAPE UP       |
| WARD          | TKB SPORT SHOP |
+---------------+----------------+
9 rows in set (0.06 sec)
```

6. *Display salesman who did not earn commission, with customer names whom they are dealing with.*

```
mysql>  select s.ename as salesman_name,
        c.name as customer_name
from emp s
join customer c
on s.empno = c.repid
where s.job = 'salesman'
  and (s.comm is null or s.comm = 0)
order by s.ename, c.name;
+---------------+---------------+
| salesman_name | customer_name |
+---------------+---------------+
| TURNER        | JOCKSPORTS    |
| TURNER        | K + T SPORTS  |
| TURNER        | NORTH WOODS   |
+---------------+---------------+
3 rows in set (0.05 sec)
```

7. *Display salesman names, salary, commission and their salary grades.*

```
mysql> select s.ename as salesman_name,
       s.sal as salary,
       s.comm as commission,
       g.grade as salary_grade
from emp s
join salgrade g
on s.sal between g.losal and g.hisal
where s.job = 'salesman'
order by s.ename;
+---------------+--------+------------+--------------+
| salesman_name | salary | commission | salary_grade |
+---------------+--------+------------+--------------+
| AARUSH        |   1350 |       2700 |            2 |
| ALLEN         |   1600 |        300 |            3 |
| GRASS         |   2575 |       2700 |            4 |
| MARTIN        |   1250 |       1400 |            2 |
| TURNER        |   1500 |          0 |            3 |
| WARD          |   1250 |        500 |            2 |
+---------------+--------+------------+--------------+
6 rows in set (0.00 sec)
```

8. *Display all Tennis products and their price details (standard and minimum price).*

```
myql> select p.descrip as product_name,
      p.price as standard_price,
      (select min(p2.price)
       from product p2
       where p2.descrip = p.descrip) as minimum_price
from product p
where p.descrip like '%tennis%'
order by p.descrip, p.price;
```

```
+-------------------------+----------------+---------------+
| product_name            | standard_price | minimum_price |
+-------------------------+----------------+---------------+
| ACE TENNIS BALLS-3 PACK |            120 |           120 |
| ACE TENNIS BALLS-3 PACK |            125 |           120 |
| ACE TENNIS BALLS-3 PACK |            500 |           120 |
| ACE TENNIS BALLS-6 PACK |            100 |           100 |
| ACE TENNIS BALLS-6 PACK |            155 |           100 |
| ACE TENNIS BALLS-6 PACK |            800 |           100 |
| ACE TENNIS NET          |            150 |           150 |
| ACE TENNIS NET          |            400 |           150 |
| ACE TENNIS RACKET I     |            110 |           110 |
| ACE TENNIS RACKET I     |            180 |           110 |
| ACE TENNIS RACKET II    |            105 |           105 |
| ACE TENNIS RACKET II    |            190 |           105 |
| RH: "GUIDE TO TENNIS"   |            145 |           145 |
| SP TENNIS RACKET        |            110 |           110 |
| SP TENNIS RACKET        |            345 |           110 |
+-------------------------+----------------+---------------+
15 rows in set (0.04 sec)
```

9. *Display the products purchased by either JOCKSPORTS (customer number 100) or JUST TENNIS (customer number 103) or both of them.*

```
Mysql> select c.name as customer_name,
       p.descrip as product_name,
       p.price as product_price
from customer c
join orders o
on c.custid = o.cnum
join product p
on o.onum = p.prodid
where c.custid in (100, 103)
order by c.name, p.descrip;
Empty set (0.00 sec)
```

10. *Display the department name, and salary of any employee whose salary and commission matches both the salary and commission of any employee in DALLAS.*

```
select d.dname as department_name,
       e.sal as salary,
       e.comm as commission
from emp e join dept d on e.deptno = d.deptno
where (e.sal, e.comm) in (
   select e2.sal, e2.comm
   from emp e2
   join dept d2
   on e2.deptno = d2.deptno
   where d2.loc = 'dallas'
) order by d.dname, e.sal;

Empty set (0.04 sec)
```

11. *List out the common jobs in Research and Accounting Departments in ascending order.*

```
mysql> select distinct e.job
from emp e
join dept d
on e.deptno = d.deptno
where d.dname = 'research'
 and e.job in (
    select e2.job
    from emp e2
    join dept d2
    on e2.deptno = d2.deptno
    where d2.dname = 'accounting'
 )
order by e.job;

+----------+---------+
| job                |
+-----------------+
| CLERK          |
| MANAGER   |
+----------+
2 rows in set (0.00 sec)
```